

Infrastruktura (jednorazowo)

```
-- services.msc:
-- SQL Server (MSSQLSERVER) -> Lokalne konto systemowe -> Restart
-- SQL Server Browser -> Automatic -> Start
-- SQL Server Configuration Manager (Admin):
-- Network Conf. -> MSSQLSERVER: TCP/IP Enable, Named Pipes Enable
-- SQL Server Services -> Restart MSSQLSERVER
-- Firewall: Reguly wejsciowe + wyjsciowe -> Port TCP 1433 (SQL), 1521 (Oracle)
-- Oracle Net Manager: Local -> Service Naming -> Create
-- Name=PD25, Protocol=TCP, Host=212.51.216.169, Port=1521, SID=PD25
-- Sterownik: Microsoft.ACE.OLEDB.12.0 (1ub 16.0)
-- Plik C:\Northwind -> Properties -> Security -> Add Everyone (Read+Execute)
```

Konfiguracja SQL Server

```
EXEC sp_configure 'show advanced options', 1; RECONFIGURE;
EXEC sp_configure 'Ad Hoc Distributed Queries', 1; RECONFIGURE;
GO
USE master;
EXEC master.dbo.sp_MSset_oledb_prop N'MSOLEDBSQL', N'AllowInProcess', 1;
EXEC master.dbo.sp_MSset_oledb_prop N'MSOLEDBSQL', N'DynamicParameters', 1;
EXEC master.dbo.sp_MSset_oledb_prop N'OracleDB.Oracle', N'AllowInProcess', 1;
EXEC master.dbo.sp_MSset_oledb_prop N'OracleDB.Oracle', N'DynamicParameters', 1;
EXEC master.dbo.sp_MSset_oledb_prop N'Microsoft.ACE.OLEDB.12.0', N'AllowInProcess', 1;
EXEC master.dbo.sp_MSset_oledb_prop N'Microsoft.ACE.OLEDB.12.0', N'DynamicParameters', 1;
GO
```

Login / User / Granty

```
CREATE LOGIN KONT01 WITH PASSWORD = '12345', CHECK_POLICY = OFF;
USE Northwind;
CREATE USER KONT01 FOR LOGIN KONT01;
GRANT SELECT, INSERT, UPDATE, DELETE ON SCHEMA::dbo TO KONT01;
GO
```

Linked server: SQL → SQL (Serwer1, sa → KONT01)

```
EXEC sp_addlinkedserver @server='Serwer1', @srvproduct='', @provider='MSOLEDBSQL',
    @datasrc='WB-20', @catalog='Northwind';
EXEC sp_addlinkedsrvlogin @rmtsrvname='Serwer1', @useself='false',
    @locallogin='sa', @rmtuser='KONT01', @rmtpassword='12345';
EXEC sp_serveroption 'Serwer1', 'data access','true';
EXEC sp_serveroption 'Serwer1', 'rpc','true';
EXEC sp_serveroption 'Serwer1', 'rpc out','true';
GO
```

Linked server: SQL → Oracle

```
EXEC sp_addlinkedserver @server='OraPD25', @srvproduct='Oracle',
    @provider='OracleDB.Oracle',
    @datasrc='(DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=212.51.216.169)(PORT=1521))(CONNECT_DATA=(
        SID=PD25)))';
EXEC sp_addlinkedsrvlogin @rmtsrvname='OraPD25', @useself='false',
    @locallogin='sa', @rmtuser='PD251190', @rmtpassword='12345';
EXEC sp_serveroption 'OraPD25','rpc out','true';
GO
```

Linked server: SQL → Access / Excel

```
EXEC sp_addlinkedserver @server='AccessNW', @srvproduct='OLE DB Provider for ACE',
    @provider='Microsoft.ACE.OLEDB.12.0', @datasrc='C:\Northwind\Northwind.mdb';
EXEC sp_addlinkedserver @server='ExcelST', @srvproduct='OLE DB Provider for ACE',
    @provider='Microsoft.ACE.OLEDB.12.0', @datasrc='C:\Northwind\listy.xlsx',
    @provstr='Excel 12.0;HDR=YES';
GO
```

OPENROWSET: SQL / Oracle / Access / Excel

```
SELECT a.* FROM OPENROWSET('MSOLEDBSQL',
    'Server=WB-20;Database=Northwind;UID=sa;PWD=praktyka;',
    'SELECT ProductID, ProductName FROM dbo.Products') AS a;

SELECT a.* FROM OPENROWSET('OracleDB.Oracle',
    '(DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=212.51.216.169)(PORT=1521))(CONNECT_DATA=(SID=PD25)))',
    'PD251190','12345',
    'SELECT EmployeeID, LastName, FirstName FROM Employees') AS a;

-- Access: ZAWSZE SELECT *, JOIN/WHERE poza OPENROWSET
SELECT a.* FROM OPENROWSET('Microsoft.ACE.OLEDB.12.0',
    'C:\Northwind\Northwind.mdb','admin','',
    'SELECT * FROM [Opisy zamowien]') AS a;

SELECT a.* FROM OPENROWSET('Microsoft.ACE.OLEDB.12.0',
    'Excel 12.0;Database=C:\Northwind\listy.xlsx;HDR=YES;',
    'SELECT * FROM [oceny_do_www$]') AS a;
```

OPENROWSET multi-zrodlo (CTE)

```
;WITH klienci AS (
    SELECT * FROM OPENROWSET('OracleDB.Oracle',
        '(DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=212.51.216.169)(PORT=1521))(CONNECT_DATA=(SID=PD25))
        )','PD251190','12345',
        'SELECT CustomerID, CompanyName FROM Customers')),
zamowienia AS (
    SELECT * FROM OPENROWSET('MSOLEDBSQL',
        'Server=WB-20;Database=Northwind;UID=sa;PWD=praktyka;',
        'SELECT OrderID, CustomerID FROM dbo.Orders')),
produkty AS (
    SELECT * FROM OPENROWSET('Microsoft.ACE.OLEDB.12.0',
        'C:\Northwind\Northwind.mdb','admin','',
        'SELECT * FROM Produkty'))
SELECT k.CompanyName, z.OrderID, p.NazwaProduktu
FROM zamowienia z
JOIN klienci k ON z.CustomerID = k.CustomerID
JOIN Northwind.dbo.[Order Details] od ON z.OrderID = od.OrderID
JOIN produkty p ON od.ProductID = p.IDproduktu;
```

OPENQUERY (SELECT/INSERT/UPDATE/DELETE/EXEC AT)

```
SELECT * FROM OPENQUERY(Serwer1, 'SELECT ProductID, ProductName FROM Northwind.dbo.Products');
INSERT INTO OPENQUERY(Serwer1, 'SELECT ProductName, UnitPrice FROM Northwind.dbo.Products') VALUES
('Test', 1.0);
UPDATE OPENQUERY(Serwer1, 'SELECT ProductName, UnitPrice FROM Northwind.dbo.Products WHERE
    ProductName = ''Test'') SET UnitPrice = 9.99;
DELETE FROM OPENQUERY(Serwer1, 'SELECT ProductID FROM Northwind.dbo.Products WHERE ProductName = ''
    Test'');
EXEC ('SELECT TOP 5 * FROM Northwind.dbo.Products') AT Serwer1;
```

4-czlonowa identyfikacja

```
SELECT TOP 5 * FROM Serwer1.Northwind.dbo.Products;
SELECT TOP 5 * FROM Serwer1.Northwind.dbo.[Order Details];
```

Widok zaszyfrowany z OPENROWSET

```
CREATE OR ALTER VIEW dbo.Widok1 WITH ENCRYPTION AS
SELECT e.EmployeeID, e.LastName, o.OrderID, p.ProductName
FROM OPENROWSET('OracleDB.Oracle',
    '(DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=212.51.216.169)(PORT=1521))(CONNECT_DATA=(SID=PD25)))',
    'PD251190','12345',
    'SELECT EmployeeID, LastName FROM Employees') AS e
INNER JOIN OPENROWSET('OracleDB.Oracle',
    '(DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=212.51.216.169)(PORT=1521))(CONNECT_DATA=(SID=PD25)))',
    'PD251190','12345',
    'SELECT OrderID, EmployeeID FROM Orders') AS o ON e.EmployeeID = o.EmployeeID
INNER JOIN OPENROWSET('MSOLEDBSQL',
    'Server=WB-20;Database=Northwind;UID=sa;PWD=praktyka;',
    'SELECT ProductID, ProductName FROM dbo.Products') AS p ON e.i=i;
GO
SELECT definition FROM sys.sql_modules WHERE object_id = OBJECT_ID('dbo.Widok1');
```

Procedura z parametrem (Access, CTE)

```
CREATE OR ALTER PROCEDURE dbo.PROC4 @OrderID INT AS
WITH o AS (SELECT * FROM OPENROWSET('Microsoft.ACE.OLEDB.12.0',
    'C:\Northwind\Northwind.mdb','admin','', 'SELECT * FROM [Opisy zamowien]')),
p AS (SELECT * FROM OPENROWSET('Microsoft.ACE.OLEDB.12.0',
    'C:\Northwind\Northwind.mdb','admin','', 'SELECT * FROM Produkty')),
k AS (SELECT * FROM OPENROWSET('Microsoft.ACE.OLEDB.12.0',
    'C:\Northwind\Northwind.mdb','admin','', 'SELECT * FROM Kategorie'))
SELECT k.NazwaKategorii AS Kategoria, SUM(o.CenaJednostkowa * o.Ilosc) AS WartoscBezRabatu
FROM o JOIN p ON o.IDproduktu = p.IDproduktu
JOIN k ON p.IDkategorii = k.IDkategorii
WHERE o.IDzamowienia = @OrderID
GROUP BY k.NazwaKategorii;
GO
EXEC dbo.PROC4 10248;
EXEC dbo.PROC4 @OrderID = 10250;
DECLARE @id INT = 10251; EXEC dbo.PROC4 @id;
DECLARE @id INT = 10252; EXEC dbo.PROC4 @OrderID = @id;
```

DTC (rozproszona transakcja)

```
SET XACT_ABORT ON;
BEGIN DISTRIBUTED TRANSACTION;
    INSERT INTO Northwind.dbo.Categories (CategoryName, Description) VALUES ('LokalnaTest','lokalna')
    ;
    INSERT INTO OPENQUERY(Serwer1, 'SELECT CategoryName, Description FROM Northwind.dbo.Categories')
        VALUES ('ZdalnaTest','zdalna');
COMMIT TRANSACTION;
```

Oracle: konto (po stronie SYS)

```
-- CREATE USER NORTHWIND IDENTIFIED BY '123456';
-- GRANT CONNECT, RESOURCE, UNLIMITED TABLESPACE TO NORTHWIND;
-- GRANT CREATE SESSION, CREATE TABLE, CREATE VIEW TO NORTHWIND;
```

Przydatne sp_*

```
EXEC sp_linkedservers;
EXEC sp_helpserver;
EXEC sp_tables_ex 'Serwer1';
EXEC sp_columns_ex 'Serwer1','Products';
EXEC sp_dropserver 'Serwer1','droplogins';
```

Sprzatanie / DROP / ALTER

```
DROP VIEW IF EXISTS dbo.Widok1;
DROP PROCEDURE IF EXISTS dbo.PROC4;
EXEC sp_dropserver 'Serwer1','droplogins'; -- usuwa linked + mapping
DROP USER IF EXISTS KONT01; -- najpierw user w bazie
DROP LOGIN KONT01; -- potem login
ALTER LOGIN sa WITH PASSWORD = 'stare';
ALTER LOGIN KONT01 DISABLE; -- wytlacz konto
```

Paterny zadan kolekwyjnych

```
-- A) Wzrost YoY (rok do roku) - kto największy wzrost
;WITH r1997 AS (
    SELECT s.ShipperID, SUM(od.UnitPrice*od.Quantity) AS W
    FROM Northwind.dbo.Shippers s
    JOIN Northwind.dbo.Orders o ON s.ShipperID = o.ShipVia
    JOIN Northwind.dbo.[Order Details] od ON o.OrderID = od.OrderID
    WHERE YEAR(o.OrderDate) = 1997 GROUP BY s.ShipperID),
r1998 AS (... analogicznie YEAR=1998 ...)
SELECT TOP 1 a.ShipperID, b.W - a.W AS Wzrost
FROM r1997 a JOIN r1998 b ON a.ShipperID = b.ShipperID
ORDER BY Wzrost DESC;

-- B) Kto nie zrealizował (NOT EXISTS lokalnie + OPENQUERY zdalnie)
SELECT c.CustomerID FROM Northwind.dbo.Customers c
WHERE NOT EXISTS (SELECT 1 FROM Northwind.dbo.Orders o WHERE o.CustomerID = c.CustomerID);

SELECT * FROM OPENQUERY(Serwer1, '
    SELECT c.CustomerID FROM Northwind.dbo.Customers c
    LEFT JOIN Northwind.dbo.Orders o ON c.CustomerID = o.CustomerID
    GROUP BY c.CustomerID HAVING COUNT(o.OrderID)=0');

-- C) TOP N po agregacji (największy klient/produkt)
SELECT TOP 3 p.ProductName, SUM(od.UnitPrice*od.Quantity) AS W
FROM Northwind.dbo.Products p
JOIN Northwind.dbo.[Order Details] od ON p.ProductID = od.ProductID
GROUP BY p.ProductName ORDER BY W DESC;

-- D) Suma sprzedaży w roku (procedura przez linked server)
CREATE OR ALTER PROCEDURE dbo.SprzedazRok @Rok INT AS
SELECT YEAR(o.OrderDate) AS Rok,
    SUM(od.UnitPrice * od.Quantity * (1 - od.Discount)) AS Wartosc
FROM Serwer1.Northwind.dbo.Orders o
JOIN Serwer1.Northwind.dbo.[Order Details] od ON o.OrderID = od.OrderID
WHERE YEAR(o.OrderDate) = @Rok GROUP BY YEAR(o.OrderDate);
GO
GRANT EXECUTE ON dbo.SprzedazRok TO KONT01;

-- E) Zapytanie rozproszone 4 zrodla (Customers lokal + Orders/OD/Products zdalne)
SELECT c.CompanyName, o.OrderID, p.ProductName, od.UnitPrice*od.Quantity AS W
FROM Northwind.dbo.Customers c
JOIN Serwer1.Northwind.dbo.Orders o ON c.CustomerID = o.CustomerID
JOIN Serwer1.Northwind.dbo.[Order Details] od ON o.OrderID = od.OrderID
JOIN Serwer1.Northwind.dbo.Products p ON od.ProductID = p.ProductID;
```

INSERT/DELETE przez linked + OPENQUERY (lokalny SQL i Oracle)

```
-- 4-czlonowo
INSERT INTO Serwer1.Northwind.dbo.Products (ProductName, UnitPrice) VALUES ('TestX', 1);
DELETE FROM Serwer1.Northwind.dbo.Products WHERE ProductName='TestX';

-- OPENQUERY (T-SQL wykonywany po zdalnej stronie)
```

```
INSERT INTO OPENQUERY(Server1,'SELECT ProductName, UnitPrice FROM Northwind.dbo.Products')
VALUES ('TestY', 2);
```

```
-- Oracle przez OPENQUERY
INSERT INTO OPENQUERY(OraPD25,'SELECT EMPLOYEEID, LASTNAME, FIRSTNAME FROM EMPLOYEES')
VALUES (999, 'Test', 'X');
DELETE FROM OPENQUERY(OraPD25,'SELECT EMPLOYEEID FROM EMPLOYEES WHERE EMPLOYEEID=999');
```

Tabela docelowa (SQL i Oracle)

```
-- SQL Server
CREATE TABLE dbo.koledzy (
    indeks INT NOT NULL PRIMARY KEY,
    nazwisko VARCHAR(50) NOT NULL,
    imie VARCHAR(25) NOT NULL;
GRANT SELECT, INSERT, UPDATE, DELETE ON dbo.koledzy TO PUBLIC;
```

```
-- Oracle (po stronie NORTHWIND/PD251190 w SQL Developer)
-- CREATE TABLE KOLEDZY (
--     INDEKS NUMBER(16) NOT NULL PRIMARY KEY,
--     NAZWISKO VARCHAR2(50) NOT NULL,
--     IMIE VARCHAR2(25) NOT NULL;
-- GRANT SELECT, INSERT, UPDATE, DELETE ON KOLEDZY TO PUBLIC;
```

Procedura SQL z parametrami (insert lokalny)

```
CREATE OR ALTER PROCEDURE dbo.dodaj_kolege_sql
    @indeks INT,
    @nazwisko VARCHAR(50),
    @imie VARCHAR(25)
AS
BEGIN
    SET NOCOUNT ON;
    INSERT INTO dbo.koledzy (indeks, nazwisko, imie)
    VALUES (@indeks, @nazwisko, @imie);
END;
GO
EXEC dbo.dodaj_kolege_sql @indeks=62001, @nazwisko='Nowak', @imie='Anna';
```

Procedura Oracle (PL/SQL, po stronie Oracle)

```
-- CREATE OR REPLACE PROCEDURE DODAJ_KOLEGE_ORACLE (
--     P_INDEKS IN NUMBER,
--     P_NAZWISKO IN VARCHAR2,
--     P_IMIE IN VARCHAR2)
-- AS BEGIN
--     INSERT INTO KOLEDZY (INDEKS, NAZWISKO, IMIE)
--     VALUES (P_INDEKS, P_NAZWISKO, P_IMIE);
-- END;
-- /
```

DTC: TRY/CATCH + DISTRIBUTED TRANSACTION (lokal + OPENROWSET Oracle)

```
SET XACT_ABORT ON;
GO
BEGIN TRY
    BEGIN DISTRIBUTED TRANSACTION;

    INSERT INTO dbo.koledzy (indeks, nazwisko, imie)
    VALUES (61001, 'Kowalski', 'Jan');

    INSERT INTO OPENROWSET(
        'OraOLEDB.Oracle',
        '(DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=212.51.216.169)(PORT=1521))(CONNECT_DATA=(SID=PD25))
        );'PD251190','12345',
        'SELECT INDEKS, NAZWISKO, IMIE FROM KOLEDZY')
    VALUES (61001, 'Kowalski', 'Jan');

    COMMIT TRANSACTION;
END TRY
BEGIN CATCH
    IF @@TRANCOUNT > 0 ROLLBACK TRANSACTION;
    THROW;
END CATCH;
GO
```

DTC: TRY/CATCH + procedura SQL + EXEC AT Oracle

```
SET XACT_ABORT ON;
GO
BEGIN TRY
    BEGIN DISTRIBUTED TRANSACTION;

    EXEC dbo.dodaj_kolege_sql @indeks=62001, @nazwisko='Nowak', @imie='Anna';

    EXEC ('BEGIN DODAJ_KOLEGE_ORACLE(62001, ''Nowak'', ''Anna''); END;') AT OraPD25;

    COMMIT TRANSACTION;
END TRY
BEGIN CATCH
    IF @@TRANCOUNT > 0 ROLLBACK TRANSACTION;
    THROW;
END CATCH;
GO
```

Default mapping (locallogin = NULL)

```
-- mapuje WSZYSTKIE lokalne loginy na ten sam zdalny user
EXEC sp_addlinkedsvlogin 'Server1','False', NULL, 'KONT01','12345';
EXEC sp_addlinkedsvlogin 'OraPD25','False', NULL, 'PD251190','12345';
```